STAFF REVIEW · H446 091 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. CIR.

2. Stores address being accessed.

3. PC.

4. $1+0+1=10_2$, so Sum0, carry-out1.

I Do — source worked steps

1. Explain the mechanism — Fetch: MAR30, CIR ADD70, PC31. Execution reads memory[70]=5 and ALU adds to ACC4, giving ACC9. — The PC increments in fetch while the ALU operates on ACC and fetched operand.

We Do — guided CHECK

1. Explain the mechanism
After fetch PC43,CIR SUB90. Operand read from90 into MDR=3; ALU computes11−3, ACC8.
The addressed operand must not be confused with instruction address42.

You Do — source mark schemes

1. MAR20 [2], memory[20] via MDR [2], CIR gets instruction [1], PC21 [1].

2. CU interprets opcode/operand [2], memory operand address80 [1], data13 via MDR [1], ACC13 [2].

3. MAR80 carries target address [2], memory sends value13 on data path [2], MDR receives13 [2].

Exit Ticket — source answer

1. True. MAR holds the address; MDR the returned data/instruction.